L5: Dot Products, Orthonormal Bases and Orthogonal Projection

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Linear Algebra in Machine Learning
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In the previous lecture, we saw that any k linearly independent vectors in a k-dimensional subspace of $\mathbb{R}^d$ form a basis. Relative to this basis, every vector in the subspace has a unique coordinate vector in $\mathbb{R}^k$. Therefore, if the datapoints in a dataset lie in such a subspace and $k \ll d$, we can "compress" each datapoint from a vector in $\mathbb{R}^d$ to one in $\mathbb{R}^k$. These lower-dimensional representations can reduce storage needs and the computation required by subsequent learning tasks. This lecture develops the linear algebra tools needed to make this process useful in practice: dot products, orthonormal bases, orthogonal matrices and orthogonal projection.

Dot products measure alignment

In Lecture 1, we represented English words by *embedding* vectors. Search queries and documents can also be represented by vectors. We can then use these vector representations to measure similarity or alignment between objects.

Example 1. Suppose a search query and two documents are represented by term-frequency vectors whose coordinates record how often each of three fixed words appears in the query or each document:

$$\mathbf{q} = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}, \qquad \mathbf{d}_1 = \begin{pmatrix} 1 \\ 8 \\ 2 \end{pmatrix}, \qquad \mathbf{d}_2 = \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix}.$$

One way to measure the similarity or alignment between two such vectors is to compute their *dot product*. Multiplying corresponding entries and adding gives the scores

$$4 \times 1 + 1 \times 8 + 3 \times 2 = 18 \qquad \text{and} \qquad 4 \times 6 + 1 \times 2 + 3 \times 5 = 41.$$

The second document has higher counts for the first and third words, which the query emphasizes. Under this dot-product scoring rule, it receives the higher score and would be ranked first.

Definition 2. For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, their dot product is

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\mathsf{T}\mathbf{y} = \sum_{j=1}^{d} x_j y_j.$$

In particular, $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$.

The dot product is symmetric and linear in each input. For example, $\langle \mathbf{x}, a\mathbf{y} + b\mathbf{z} \rangle = a \langle \mathbf{x}, \mathbf{y} \rangle + b \langle \mathbf{x}, \mathbf{z} \rangle$. These properties follow directly by expanding the coordinate sums.

A raw count does not distinguish a word that is central to an article from one that appears incidentally in a much longer article. For example, suppose a query contains only the word "cat". One article mentions "cat" twice in 100 words, while another mentions it five times in 10,000 words. The raw dot-product score

is higher for the longer article, even though "cat" is more prominent in the shorter one. This motivates seeking a normalized measure of vector similarity. One common choice is cosine similarity, which divides the dot product by the Euclidean lengths of the term-frequency vectors and thus ignores their overall scale. The following inequality guarantees that this normalized score lies between -1 and 1.

Theorem 3. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \;\leq\; \|\mathbf{x}\|\|\mathbf{y}\|.$$

Equality holds exactly when $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof: If $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$, both sides are zero and the vectors are linearly dependent. Now suppose $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{y} \neq \mathbf{0}$. Since a squared norm is nonnegative, for any real number a,

$$\begin{aligned} 0 \;\leq\; \|\mathbf{x} - a\mathbf{y}\|^2 &= \langle \mathbf{x} - a\mathbf{y}, \mathbf{x} - a\mathbf{y} \rangle \\ &= \langle \mathbf{x}, \mathbf{x} \rangle - a \langle \mathbf{y}, \mathbf{x} \rangle - a \langle \mathbf{x}, \mathbf{y} \rangle + a^2 \langle \mathbf{y}, \mathbf{y} \rangle \\ &= \|\mathbf{x}\|^2 - 2a \langle \mathbf{x}, \mathbf{y} \rangle + a^2 \|\mathbf{y}\|^2. \end{aligned}$$

Set $a = \langle \mathbf{x}, \mathbf{y} \rangle / \|\mathbf{y}\|^2$. The inequality becomes

$$0 \;\leq\; \|\mathbf{x}\|^2 - \frac{\langle \mathbf{x}, \mathbf{y} \rangle^2}{\|\mathbf{y}\|^2}.$$

Multiplying by $\|\mathbf{y}\|^2$ gives

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \;\leq\; \|\mathbf{x}\|^2 \|\mathbf{y}\|^2,$$

and taking square roots proves the inequality. For this choice of a, equality holds exactly when $\|\mathbf{x} - a\mathbf{y}\| = 0$, equivalently when $\mathbf{x} = a\mathbf{y}$. □

Definition 4. For nonzero vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, the angle $\theta \in [0, \pi]$ between them is defined by

$$\cos\theta \;=\; \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

The value on the right is their cosine similarity. Any vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, including zero vectors, are orthogonal, written $\mathbf{x} \perp \mathbf{y}$, if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

Example 5. Return to the query and documents in Example 1. Their cosine similarities are

$$\frac{\langle \mathbf{q}, \mathbf{d}_1 \rangle}{\|\mathbf{q}\|\|\mathbf{d}_1\|} \;=\; \frac{18}{\sqrt{26}\sqrt{69}} \approx 0.425, \qquad \frac{\langle \mathbf{q}, \mathbf{d}_2 \rangle}{\|\mathbf{q}\|\|\mathbf{d}_2\|} \;=\; \frac{41}{\sqrt{26}\sqrt{65}} \approx 0.997.$$

Unlike the dot product, cosine similarity removes the effect of vector length and compares only directions. Under this cosine similarity scoring rule, the second document points almost exactly in the query's direction and should be ranked first.

Orthonormal bases make coordinates easy

In Lecture 4, we defined a vector's coordinates relative to an ordered basis. We now collect the basis vectors into a matrix to compute these coordinates.

Definition 6. Let $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_k)$ be an ordered basis of a subspace $\mathcal{S} \subseteq \mathbb{R}^d$. Its basis matrix is

$$\mathbf{B} \;=\; \begin{pmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_k \\ | & & | \end{pmatrix} \;\in\; \mathbb{R}^{d\times k}.$$

For every $\mathbf{x} \in \mathcal{S}$, we have $\mathbf{x} = \mathbf{B}[\mathbf{x}]_{\mathcal{B}}$.

Given $\mathbf{x} \in \mathcal{S}$, finding its coordinate vector $\mathbf{y} = [\mathbf{x}]_{\mathcal{B}}$ amounts to solving $\mathbf{By} = \mathbf{x}$.

Since $\mathcal{B}$ is an ordered basis, the columns of $\mathbf{B}$ are linearly independent. If $\mathcal{S} = \mathbb{R}^d$, then $k = d$ and $\mathbf{B}$ is a square matrix in $\mathbb{R}^{d\times d}$. By the following proposition, $\mathbf{B}$ is invertible, so the coordinate vector of $\mathbf{x}$ can be computed as $\mathbf{y} = \mathbf{B}^{-1}\mathbf{x}$.

Proposition 7. If a square matrix $\mathbf{B} \in \mathbb{R}^{d\times d}$ has linearly independent columns, then $\mathbf{B}$ is invertible.

Proof: The columns of $\mathbf{B}$ are linearly independent exactly when the homogeneous system $\mathbf{Bz} = \mathbf{0}$ has only the trivial solution. Thus Gaussian elimination leaves no free variable, so a row-echelon form of $\mathbf{B}$ has a pivot in every one of its d columns. Since $\mathbf{B}$ also has d rows, it has a pivot in every row. Its reduced row-echelon form is therefore $\mathbf{I}_d$. By the Gauss-Jordan criterion from Lecture 3, $\mathbf{B}$ is invertible. □

A subspace can have many different bases. Orthonormal bases are particularly useful because their coordinates can be recovered directly using dot products.

Definition 8. Vectors $\mathbf{q}_1, \ldots, \mathbf{q}_k \in \mathbb{R}^d$ are orthonormal if

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle \;=\; \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Proposition 9. Every orthonormal list of vectors is linearly independent.

Proof: Suppose $\mathbf{q}_1, \ldots, \mathbf{q}_k$ are orthonormal and $\alpha_1\mathbf{q}_1 + \cdots + \alpha_k\mathbf{q}_k = \mathbf{0}$. For any $\ell \in \{1, \ldots, k\}$, take the dot product with $\mathbf{q}_\ell$ to obtain

$$0 \;=\; \langle \mathbf{q}_\ell, \mathbf{0} \rangle \;=\; \left\langle \mathbf{q}_\ell, \sum_{i=1}^{k} \alpha_i \mathbf{q}_i \right\rangle \;=\; \sum_{i=1}^{k} \alpha_i \langle \mathbf{q}_\ell, \mathbf{q}_i \rangle \;=\; \alpha_\ell.$$

Thus $\alpha_\ell = 0$ for every ℓ, so the vectors are linearly independent. □

Definition 10. Let $\mathcal{S}$ be a subspace of $\mathbb{R}^d$. An orthonormal basis of $\mathcal{S}$ is an orthonormal list of vectors whose span is $\mathcal{S}$.

Proposition 11. Let $\mathcal{Q} = (\mathbf{q}_1, \ldots, \mathbf{q}_k)$ be an orthonormal basis of a subspace $\mathcal{S}$, and let

$$\mathbf{Q} \;=\; \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_k \\ | & & | \end{pmatrix}$$

be the matrix obtained by stacking column vectors $\mathbf{q}_1, \ldots, \mathbf{q}_k$ horizontally. For every $\mathbf{x} \in \mathcal{S}$, its coordinate vector relative to $\mathcal{Q}$ is

$$[\mathbf{x}]_{\mathcal{Q}} \;=\; \begin{pmatrix} \langle \mathbf{q}_1, \mathbf{x} \rangle \\ \vdots \\ \langle \mathbf{q}_k, \mathbf{x} \rangle \end{pmatrix} \;=\; \mathbf{Q}^{\mathsf{T}}\mathbf{x}.$$

Proof: Since $\mathcal{Q}$ is a basis of $\mathcal{S}$, there are unique coefficients $\alpha_1, \ldots, \alpha_k$ such that

$$\mathbf{x} \;=\; \sum_{i=1}^{k} \alpha_i \mathbf{q}_i.$$

For any $j \in \{1, \ldots, k\}$, take the dot product with $\mathbf{q}_j$ to obtain

$$\langle \mathbf{q}_j, \mathbf{x} \rangle \;=\; \sum_{i=1}^{k} \alpha_i \langle \mathbf{q}_j, \mathbf{q}_i \rangle \;=\; \alpha_j.$$

Thus the j-th coordinate of $\mathbf{x}$ relative to $\mathcal{Q}$ is $\langle \mathbf{q}_j, \mathbf{x} \rangle$. Since this holds for every j, stacking the coordinates gives $[\mathbf{x}]_{\mathcal{Q}} = \mathbf{Q}^{\mathsf{T}}\mathbf{x}$. □

Thus coordinates in an orthonormal basis require only dot products; there is no need to solve a linear system or compute a matrix inverse.

If $\mathcal{S} = \mathbb{R}^d$, an orthonormal basis has d vectors, and $\mathbf{Q}$ is a square matrix. Orthonormality implies that $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_d$. Such square matrices have a special name and several useful properties.

Definition 12. A square matrix $\mathbf{Q} \in \mathbb{R}^{d \times d}$ is an orthogonal matrix if its columns form an orthonormal basis of $\mathbb{R}^d$. Equivalently,

$$\mathbf{Q}^{\mathsf{T}}\mathbf{Q} \;=\; \mathbf{I}_d.$$

Proposition 13. If $\mathbf{Q} \in \mathbb{R}^{d \times d}$ is orthogonal, then

$$\mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}, \qquad \langle \mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle, \qquad \left\langle \mathbf{Q}^{\mathsf{T}}\mathbf{x}, \mathbf{Q}^{\mathsf{T}}\mathbf{y} \right\rangle = \langle \mathbf{x}, \mathbf{y} \rangle, \qquad \|\mathbf{Q}\mathbf{x}\| \;=\; \|\mathbf{x}\|$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$. Hence the linear map represented by $\mathbf{Q}$ preserves dot products and lengths, and it preserves angles between nonzero vectors.

Proof: The columns of $\mathbf{Q}$ are orthonormal and hence linearly independent by Proposition 9. Therefore $\mathbf{Q}$ is invertible by Proposition 7. Multiplying $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_d$ on the right by $\mathbf{Q}^{-1}$ gives $\mathbf{Q}^\mathsf{T} = \mathbf{Q}^{-1}$. Moreover,

$$\langle \mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{y} \rangle \;=\; (\mathbf{Q}\mathbf{x})^\mathsf{T}(\mathbf{Q}\mathbf{y}) \;=\; \mathbf{x}^\mathsf{T}\mathbf{Q}^\mathsf{T}\mathbf{Q}\mathbf{y} \;=\; \mathbf{x}^\mathsf{T}\mathbf{y},$$

and

$$\left\langle \mathbf{Q}^\mathsf{T}\mathbf{x}, \mathbf{Q}^\mathsf{T}\mathbf{y} \right\rangle \;=\; (\mathbf{Q}^\mathsf{T}\mathbf{x})^\mathsf{T}(\mathbf{Q}^\mathsf{T}\mathbf{y}) \;=\; \mathbf{x}^\mathsf{T}\mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{y} \;=\; \mathbf{x}^\mathsf{T}\mathbf{y}.$$

Taking $\mathbf{y} = \mathbf{x}$ gives $\|\mathbf{Q}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$, and both norms are nonnegative. The formula for cosine similarity then shows that angles between nonzero vectors are preserved. □

Gram-Schmidt constructs orthonormal bases

One question remains: does every subspace have an orthonormal basis? The answer is yes. The *Gram-Schmidt process* is one standard construction.

Theorem 14. Let $\mathbf{v}_1, \ldots, \mathbf{v}_k$ be linearly independent. Define

$$\mathbf{u}_1 \;=\; \mathbf{v}_1, \qquad \mathbf{q}_1 \;=\; \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|},$$

and, for $j = 2, \ldots, k$, define

$$\mathbf{u}_j \;=\; \mathbf{v}_j - \sum_{i=1}^{j-1} \langle \mathbf{q}_i, \mathbf{v}_j \rangle \, \mathbf{q}_i, \qquad \mathbf{q}_j \;=\; \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}.$$

Then $\mathbf{q}_1, \ldots, \mathbf{q}_k$ are orthonormal and, for every j,

$$\operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_j\} \;=\; \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_j\}.$$

Proof: We argue successively for $j = 1, \ldots, k$. Normalizing $\mathbf{u}_1 = \mathbf{v}_1 \neq \mathbf{0}$ gives a unit vector with the same span. Suppose the claim holds through $j-1$. For every $\ell < j$, orthonormality of the earlier vectors gives

$$\begin{aligned} \langle \mathbf{q}_\ell, \mathbf{u}_j \rangle \;&=\; \langle \mathbf{q}_\ell, \mathbf{v}_j \rangle - \sum_{i=1}^{j-1} \langle \mathbf{q}_i, \mathbf{v}_j \rangle \, \langle \mathbf{q}_\ell, \mathbf{q}_i \rangle \\ &=\; \langle \mathbf{q}_\ell, \mathbf{v}_j \rangle - \langle \mathbf{q}_\ell, \mathbf{v}_j \rangle \;=\; 0. \end{aligned}$$

Moreover, $\mathbf{u}_j$ is $\mathbf{v}_j$ minus a linear combination of the earlier vectors. If $\mathbf{u}_j$ were zero, then $\mathbf{v}_j$ would lie in the span of $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$, contradicting independence. Hence $\mathbf{q}_j$ is well-defined, has unit length and is orthogonal to the earlier $\mathbf{q}_i$. Finally, each of $\mathbf{u}_j$ and $\mathbf{v}_j$ is a linear combination of the other and the earlier vectors, so adjoining either one produces the same span. This completes the induction. □

Example 15. Start with three vectors in $\mathbb{R}^4$:

$$\mathbf{v}_1 \;=\; \begin{pmatrix} 2 \\ 4 \\ 4 \\ 8 \end{pmatrix}, \qquad \mathbf{v}_2 \;=\; \begin{pmatrix} -10 \\ 10 \\ -20 \\ 20 \end{pmatrix}, \qquad \mathbf{v}_3 \;=\; \begin{pmatrix} -28 \\ -16 \\ -11 \\ 58 \end{pmatrix}.$$

The first vector has norm 10, so

$$\mathbf{u}_1 = \mathbf{v}_1, \qquad \mathbf{q}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \frac{1}{5}\begin{pmatrix}1\\2\\2\\4\end{pmatrix}.$$

Remove from $\mathbf{v}_2$ its component in the $\mathbf{q}_1$ direction, then normalize:

$$\langle \mathbf{q}_1, \mathbf{v}_2 \rangle = 10,$$

$$\mathbf{u}_2 = \mathbf{v}_2 - 10\mathbf{q}_1 = \begin{pmatrix}-12\\6\\-24\\12\end{pmatrix},$$

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{5}\begin{pmatrix}-2\\1\\-4\\2\end{pmatrix}.$$

For the third vector, remove both components already accounted for:

$$\langle \mathbf{q}_1, \mathbf{v}_3 \rangle = 30, \qquad \langle \mathbf{q}_2, \mathbf{v}_3 \rangle = 40.$$

Hence

$$\mathbf{u}_3 = \mathbf{v}_3 - 30\mathbf{q}_1 - 40\mathbf{q}_2 = \begin{pmatrix}-18\\-36\\9\\18\end{pmatrix},$$

$$\mathbf{q}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \frac{1}{5}\begin{pmatrix}-2\\-4\\1\\2\end{pmatrix}.$$

The three nonzero, mutually orthogonal residuals confirm that the starting vectors are linearly independent. The resulting $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ form an orthonormal basis of their span.

Orthogonal projection extracts the explained component

We just showed how to recover coordinates when a vector lies in a given subspace. In many ML problems, however, a datapoint does not lie exactly in the subspace. We then seek the vector in the subspace that is closest to the datapoint. *Orthogonal projection* finds this vector by retaining the components along an orthonormal basis. The residual is orthogonal to every vector in the subspace.

Definition 16. Let $\mathcal{S}$ be a subspace of $\mathbb{R}^d$. An orthogonal projection of $\mathbf{x} \in \mathbb{R}^d$ onto $\mathcal{S}$ is a vector $\mathbf{p} \in \mathcal{S}$ such that $\mathbf{x} - \mathbf{p}$ is orthogonal to every vector in $\mathcal{S}$.

Theorem 17. Let $\mathbf{Q} \in \mathbb{R}^{d\times k}$ have orthonormal columns and let $\mathcal{S} = \mathrm{Col}(\mathbf{Q})$. For every $\mathbf{x} \in \mathbb{R}^d$, its unique orthogonal projection onto $\mathcal{S}$ is

$$\mathbf{p} \;=\; \mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x}.$$

Moreover, $\mathbf{p}$ is the unique vector in $\mathcal{S}$ minimizing $\|\mathbf{x}-\mathbf{y}\|$.

Proof: The vector $\mathbf{p} = \mathbf{Q}(\mathbf{Q}^{\mathsf{T}}\mathbf{x})$ lies in $\mathcal{S} = \mathrm{Col}(\mathbf{Q})$ by definition. Let $\mathbf{r} = \mathbf{x} - \mathbf{p}$. Since $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_k$,

$$\mathbf{Q}^{\mathsf{T}}\mathbf{r} \;=\; \mathbf{Q}^{\mathsf{T}}\mathbf{x} - \mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x} \;=\; \mathbf{Q}^{\mathsf{T}}\mathbf{x} - \mathbf{Q}^{\mathsf{T}}\mathbf{x} \;=\; \mathbf{0}.$$

Therefore $\mathbf{r}$ is orthogonal to every column of $\mathbf{Q}$ and hence to every vector in $\mathcal{S}$. Thus $\mathbf{p}$ is an orthogonal projection.

For any $\mathbf{y} \in \mathcal{S}$, the vector $\mathbf{p} - \mathbf{y}$ lies in $\mathcal{S}$ and is orthogonal to $\mathbf{r}$. Expanding the squared norm gives

$$\|\mathbf{x}-\mathbf{y}\|^2 \;=\; \|\mathbf{r}+(\mathbf{p}-\mathbf{y})\|^2 \;=\; \|\mathbf{r}\|^2 + \|\mathbf{p}-\mathbf{y}\|^2 \;\geq\; \|\mathbf{r}\|^2.$$

Equality holds exactly when $\mathbf{p} = \mathbf{y}$, so $\mathbf{p}$ is the unique nearest point in $\mathcal{S}$. If $\mathbf{p}'$ is another orthogonal projection, then both $\mathbf{p} - \mathbf{p}' \in \mathcal{S}$ and $\mathbf{p} - \mathbf{p}' = (\mathbf{x} - \mathbf{p}') - (\mathbf{x} - \mathbf{p})$ is orthogonal to $\mathcal{S}$. Consequently $\mathbf{p} - \mathbf{p}' = \mathbf{0}$, proving uniqueness of the orthogonal projection. □

Example 18. Let

$$\mathbf{Q} \;=\; \frac{1}{5}\begin{pmatrix} 4 & -2 & -2 \\ 2 & 4 & -1 \\ 2 & 1 & 4 \\ 1 & -2 & 2 \end{pmatrix}, \qquad \mathbf{x} \;=\; \begin{pmatrix} -8 \\ 9 \\ 3 \\ -4 \end{pmatrix}.$$

The columns of $\mathbf{Q}$ are orthonormal. Their dot products with $\mathbf{x}$ are

$$\mathbf{Q}^{\mathsf{T}}\mathbf{x} \;=\; \frac{1}{5}\begin{pmatrix} -12 \\ 63 \\ 11 \end{pmatrix}.$$

Using these dot products as coordinates relative to the columns of $\mathbf{Q}$ gives

$$\mathbf{p} \;=\; \mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x} \;=\; \frac{1}{25}\begin{pmatrix} -196 \\ 217 \\ 83 \\ -116 \end{pmatrix}.$$

The residual is given below. Multiplying it by $\mathbf{Q}^{\mathsf{T}}$ confirms that it is orthogonal to every column of $\mathbf{Q}$:

$$\mathbf{r} \;=\; \mathbf{x} - \mathbf{p} \;=\; \frac{1}{25}\begin{pmatrix} -4 \\ 8 \\ -8 \\ 16 \end{pmatrix}, \qquad \mathbf{Q}^{\mathsf{T}}\mathbf{r} \;=\; \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$